HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Administered and maintained **Water's Empower CDS**, **Biovia Smartlab**, **NuGenesis ELN**, and **Thermo Scientific LIMS** for laboratory operations, ensuring system availability and integrity.
- Served as the primary liaison with IT for **electronic system upgrades** and validation activities, coordinating testing and deployment schedules for laboratory applications.
- Managed **change controls** pertaining to electronic systems, documenting impact assessments, execution plans, and post-implementation reviews to maintain compliance.
- Provided technical guidance and training to laboratory analysts on **cGMP updates** related to electronic system usage and data integrity, improving adherence by **15%**.
- Investigated and documented non-conformances and deviations related to laboratory data, applying **Root Cause Analysis** techniques to identify and implement corrective actions.
- Developed and maintained **Python scripts** to automate data extraction and reporting from laboratory systems, reducing manual effort for batch analysis by approximately **8 hours per week**.
- Supported the qualification of **E-Systems connected analytical equipment**, ensuring all hardware and software configurations met validation requirements and local procedures.

SKILLS

Laboratory Systems – Water's Empower CDS, Biovia Smartlab, NuGenesis ELN, Thermo Scientific LIMS, Lonza MODA

Core Technologies – Python, SQL, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins

Compliance & Quality – cGMP, Computer Systems Validation (CSV), Root Cause Analysis, Change Control, Audits, Quality System Development

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, R / RStudio

Backend Development – Java, Spring Boot, REST APIs, Hibernate, JPA

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

Automated Lab Data Reporting System

- Developed a **Python-based ETL pipeline** to extract analytical test results from multiple laboratory systems (LIMS, ELN) for a financial services client.
- Engineered automated report generation using **Power BI**, creating interactive dashboards that visualized key performance indicators for sample throughput and analysis times, reducing manual reporting time by **40%**.
- Integrated the pipeline with **AWS S3** for data storage and utilized **SQLAlchemy** for efficient data manipulation across PostgreSQL and Oracle databases.

E-Systems Upgrade & Validation Support Tool

- Created a **Java Spring Boot application** with RESTful APIs to manage and track the status of electronic system upgrades and validation activities.
- Implemented a data model using **JPA and Hibernate** to store system configurations, validation documents, and test results, supporting an audit trail for regulatory compliance.
- Designed and deployed the application using **Docker** containers on an **AWS EC2** instance, ensuring scalability and ease of deployment for internal IT support teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)